

Probability and Conditional Probability

Important Concepts

1. Probability

Probability measures how likely an event is to occur.

For equally likely outcomes:

$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$

The probability always lies between **0** and **1**.

- **0** → Event is impossible
 - **1** → Event is certain
 - Between 0 and 1 → Event has some chance of occurring
-

Question 1

Probability of Drawing a Red Ball

Given

A bag contains:

- Red balls = **3**
- Blue balls = **7**

Therefore:

Total balls = $3 + 7 = 10$

Python Function

```
def calculate_probability(total_outcomes, favorable_outcomes):  
    return favorable_outcomes / total_outcomes
```

Using the Function

```
total_outcomes = 10
```

```
favorable_outcomes = 3
```

```
probability = calculate_probability(total_outcomes, favorable_outcomes)
```

```
print("Probability of drawing a red ball:", probability)
```

Output

Probability of drawing a red ball: 0.3

Calculation

$$P(\text{Red}) = \frac{3}{10} = 0.3$$

As a percentage:

$$0.3 \times 100 = 30\%$$

Final Answer

$$P(\text{Red}) = 0.3 = 30\% \quad \boxed{P(\text{Red}) = 0.3 = 30\%}$$

Therefore, the probability of drawing a red ball is **30%**.

Question 2

Simulating 20 Coin Tosses

A fair coin has two possible outcomes:

- Heads
- Tails

The theoretical probability of Heads is:

$$P(\text{Heads}) = \frac{1}{2} = 0.5$$

However, an actual experiment may produce a slightly different result.

Python Program

```
import random
```

```
heads = 0

for i in range(20):
    result = random.choice(["Heads", "Tails"])

    if result == "Heads":
        heads += 1

print("Number of Heads:", heads)

experimental_probability = heads / 20

print("Experimental probability of Heads:", experimental_probability)
```

Example Output

Because the result is random, the output can be different each time.

For example:

Number of Heads: 11
Experimental probability of Heads: 0.55

Calculation

If Heads occurs **11 times**:

$$P(\text{Heads}) = \frac{\text{Number of Heads}}{\text{Total Tosses}} = \frac{11}{20} = 0.55$$

Therefore:

$$P(\text{Heads}) = 0.55 = 55\% \quad \boxed{P(\text{Heads}) = 0.55 = 55\%}$$

Interpretation

The theoretical probability of Heads is **50%**, while the experimental probability in this particular simulation is **55%**.

The two values do not have to be exactly equal because the experiment contains only 20 tosses. With a much larger number of tosses, the experimental probability generally gets closer to the theoretical probability.

Question 3

Probability Using a Standard Six-Sided Dice

A standard die has six possible outcomes:

$\{1,2,3,4,5,6\}$

Python Function

```
def calculate_probability(total_outcomes, favorable_outcomes):  
    return favorable_outcomes / total_outcomes
```

Part A: Probability of Rolling an Even Number

The even numbers on a die are:

2, 4, 6

Therefore:

- Total outcomes = 6
- Favorable outcomes = 3

Calculation

$P(\text{Even}) = \frac{3}{6} = 0.5$

Python

```
probability_even = calculate_probability(6, 3)  
  
print("Probability of rolling an even number:", probability_even)
```

Output

Probability of rolling an even number: 0.5

Final Answer

$P(\text{Even}) = 0.5 = 50\%$

Part B: Probability of Rolling a Number Greater Than 4

Numbers greater than 4 are:

5,6,6

Therefore:

- Total outcomes = **6**
- Favorable outcomes = **2**

Calculation

$$P(\text{Number} > 4) = \frac{2}{6}$$

$$P(\text{Number} > 4) = \frac{1}{3}$$

$$P(\text{Number} > 4) \approx 0.3333$$

Python

```
probability_greater_than_4 = calculate_probability(6, 2)
```

```
print("Probability of rolling a number greater than 4:",  
      probability_greater_than_4)
```

Output

Probability of rolling a number greater than 4: 0.3333333333333333

Final Answer

$$P(\text{Number} > 4) = \frac{1}{3} \approx 33.33\%$$

Question 4

Probability Using a Deck of 52 Playing Cards

A standard deck contains:

52 cards

There are:

- 4 Aces
 - 26 red cards
 - 26 black cards
-

Part A: Probability of Drawing an Ace

There are **4 Aces** in a deck of 52 cards.

Therefore:

- Total outcomes = 52
- Favorable outcomes = 4

Calculation

$$P(\text{Ace}) = \frac{4}{52}$$

Simplifying:

$$P(\text{Ace}) = \frac{1}{13} \quad P(\text{Ace}) \approx 0.0769$$

As a percentage:

$$0.0769 \times 100 \approx 7.69\%$$

Final Answer

$$P(\text{Ace}) = \frac{1}{13} \approx 7.69\%$$

Part B: Probability of a Red Card Given That It Is a Face Card

This is a **conditional probability** problem.

Step 1: Identify the Face Cards

The question defines face cards as:

- Jack
- Queen
- King
- Ace

There are 4 types of these cards.

Since each type has 4 suits:

$$4 \times 4 = 16$$

So there are **16 face cards** under the definition given in the question.

Step 2: Count Red Face Cards

There are two red suits:

- Hearts ♥
- Diamonds ♦

Each has:

- Jack
- Queen
- King
- Ace

Therefore:

$$2 \times 4 = 8 \times 4 = 8$$

There are **8 red face cards**.

Step 3: Apply Conditional Probability

$$P(\text{Red} \mid \text{Face Card}) = \frac{\text{Red Face Cards}}{\text{Total Face Cards}} P(\text{Red} \mid \text{Face Card}) = \frac{8}{16} P(\text{Red} \mid \text{Face Card}) = 0.5$$

Final Answer

$$P(\text{Red} \mid \text{Face Card}) = 0.5 = 50\% \boxed{P(\text{Red} \mid \text{Face Card}) = 0.5 = 50\%}$$

Interpretation

If we already know that the selected card is one of the defined face cards, the probability that it is red is **50%**.

Note: In many standard card-probability questions, “face cards” means only Jack, Queen, and King, giving 12 face cards. Here, the question explicitly includes **Ace**, so we use 16 cards.

Question 5

Probability That a Pizza Order Was Placed at Night

Given Information

The food delivery app records:

$$P(\text{Night})=60\%=0.60 \quad P(\text{Day})=40\%=0.40$$

Pizza orders:

$$P(\text{Pizza} \mid \text{Night})=30\%=0.30$$

$$P(\text{Pizza} \mid \text{Day})=10\%=0.10$$

We need to find:

$$P(\text{Night} \mid \text{Pizza})$$

In words:

What is the probability that a randomly selected pizza order was placed at night?

Conditional Probability Formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Here:

- A = Night
- B = Pizza

Therefore:

$$P(\text{Night} \mid \text{Pizza}) = \frac{P(\text{Night} \cap \text{Pizza})}{P(\text{Pizza})}$$

Step 1: Calculate Night and Pizza

$$P(\text{Night} \cap \text{Pizza}) = P(\text{Night}) \times P(\text{Pizza} \mid \text{Night}) = 0.60 \times 0.30 = 0.18$$

Therefore:

$$P(\text{Night} \cap \text{Pizza}) = 0.18$$

Step 2: Calculate Total Probability of Pizza

Pizza can be ordered either during the night or during the day.

Therefore:

$$P(\text{Pizza}) = P(\text{Night})P(\text{Pizza} \mid \text{Night}) + P(\text{Day})P(\text{Pizza} \mid \text{Day})$$

Substituting the values:

$$P(\text{Pizza}) = (0.60)(0.30) + (0.40)(0.10) = 0.18 + 0.04 = 0.22$$

Therefore:

$$P(\text{Pizza})=0.22$$

Step 3: Calculate the Required Conditional Probability

$$P(\text{Night} | \text{Pizza}) = \frac{P(\text{Night} \cap \text{Pizza})}{P(\text{Pizza})} = \frac{0.18}{0.22} = 0.8182$$

As a percentage:

$$0.8182 \times 100 = 81.82\%$$

Final Answer

$$P(\text{Night} | \text{Pizza}) \approx 0.8182 = 81.82\%$$

Interpretation

There is approximately an **81.82% probability** that a randomly selected pizza order was placed at night.

This is higher than the overall percentage of night orders (60%) because pizza is ordered more frequently at night (**30%**) than during the day (**10%**).

Summary of Answers

Question	Final Answer
Q1	Probability of red ball = 0.30 or 30%
Q2	Experimental probability depends on simulation; e.g., 11 Heads → 0.55 or 55%
Q3(a)	Probability of even number = 0.50 or 50%
Q3(b)	Probability of number > 4 = 1/3 or 33.33%
Q4(a)	Probability of Ace = 1/13 or 7.69%
Q4(b)	$P(\text{Red} \text{Face Card}) = $ 0.50 or 50%
Q5	$P(\text{Night} \text{Pizza}) = $ 0.8182 or 81.82%

Key Formulas

Simple Probability

$$P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Total Probability Used in Q5

$$P(\text{Pizza}) = P(\text{Night})P(\text{Pizza} | \text{Night}) + P(\text{Day})P(\text{Pizza} | \text{Day})$$
$$= P(\text{Night})P(\text{Pizza} \mid \text{Night}) + P(\text{Day})P(\text{Pizza} \mid \text{Day})$$